

Executive Summary

CARE-LG Empirical Study — Results Report

Git SHA: 539d87b795b78bb8f9446f99b0fabeded2c84841 Seeds: [0, 1, 2, 3, 4] Generated by experiments_v2/

This report repositions CARE-LG's contribution as an empirical study answering three questions, using a 2x3(x4) ablation grid ({riemannian,euclidean} metric x {hard,soft,none} constraints, soft swept over lambda in {0.1,1,10,100}), 5 seeds per (dataset, cell), on the UCI Heart Disease (N=237 train) and NHANES-schema synthetic (N=2000 train) cohorts.

RQ1 -- Does hard pruning prevent violations that a soft-penalty Riemannian metric still commits? NO, not once you check in DECODED space. R+hard's original-space CVR is exactly 0% by construction (as designed), but its DECODED-space CVR (re-checking all constraints on the VAE's decoder output, not the original encoded features) is 83.6 ± 26.9 on UCI and 78.1 ± 10.4 on NHANES -- statistically indistinguishable from R+soft at every lambda tested (Fig D). The root cause is large VAE reconstruction error (Table: vae_reconstruction_error.csv; e.g. age off by ~6 years mean absolute error, and the binary "sex" feature reconstructed with ~0.47-0.49 mean absolute error, i.e. barely better than chance).

RQ2 -- Does the Riemannian metric change which safe path is chosen, and is it measurably better? MOSTLY NO on this architecture. R+hard and E+hard have IDENTICAL success rates on both datasets (a structural fact: hard-mode connectivity depends only on the violation predicate, not the metric). Paired Wilcoxon tests on patients where both succeeded found NO significant KDE/effort difference between R+hard and E+hard on UCI ($p=0.234$ ($n=81$ pairs, median diff=0.000) for KDE, $p=0.501$ ($n=81$ pairs, median diff=0.000) for effort) nor on NHANES ($p=0.0543$ ($n=675$ pairs, median diff=0.000) for KDE). The Riemannian pullback metric, as implemented, does not demonstrably outperform plain Euclidean latent distance under identical hard constraints on these two cohorts.

RQ3 -- When CARE-LG abstains on sparse (UCI) data, is that a certified infeasibility or a blindspot? CERTIFIED INFEASIBLE, cleanly. Across all 5 seeds, 100% of R+hard abstentions had NO constraint-satisfying low-risk target ANYWHERE in the cohort (mean 6.4 abstentions/seed out of a mean 22.6 high-risk test patients). Zero blindspots were found or resolved by either densification probe (doubled k, +500 synthetic prior samples). On NHANES, R+hard had ZERO abstentions in all 5 seeds -- there was nothing to decompose.

Bottom line: the abstention mechanism (RQ3) is CARE-LG's most defensible claim in this study. The "0% CVR" headline (RQ1) and the value-add of the Riemannian metric specifically (RQ2) are NOT supported once checked more rigorously -- see FINDINGS.md for the full, honest reconciliation with the original paper's claims.

Reproduction Check vs. Original Numbers

We reran the UNMODIFIED benchmarks/run_benchmarks.py pipeline end-to-end (fresh model training, seed=42, outputs redirected so benchmarks/ and models/ were not touched -- see experiments_v2/repro_check/).

		Committed (benchmarks/*.csv)	Reproduced (this session, same seed)
UCI	CARE-LG Success %	47.62	63.64
UCI	CARE-LG CVR %	0.00	0.00
UCI	CARE-LG KDE	-7.167	-7.238
UCI	CARE-LG Latency (s)	0.0005	0.0005
NHANES	CARE-LG Success %	99.0	98.0
NHANES	CARE-LG CVR %	0.0	0.0
NHANES	CARE-LG KDE	-8.288	-8.184
NHANES	CARE-LG Latency (s)	0.0002	0.0003

CVR and latency reproduce exactly / near-exactly -- these are structural properties of the pruned graph and the Dijkstra call. NHANES success/KDE reproduce closely (dense cohort averages out per-run training noise). UCI success rate does NOT reproduce (47.6% vs 63.6%, a 16-point swing) because neither the VAE nor the classifier is seeded anywhere in the repository (REPO_MAP.md, discrepancy D4) and UCI is small enough (N=237 train) that training-run-to-run variance dominates. Figure E in this report quantifies this directly: across 5 properly-seeded reruns of our OWN corrected pipeline, UCI R+hard success ranges from 63.6% to 87.0% (mean 71.8%, std 9.6). The paper's single point estimate of 47.6% is not a stable number and should not be reported without seed variance in future work.

Table 1 (UCI) — Full ablation grid + baselines, mean±std over 5 seeds

Method	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
R+hard (CARE-LG)	71.8±9.6	83.6±26.9	-7.392±0.213	1.72±1.38	0.0010±0.0000
R+soft ($\lambda=1$)	100.0±0.0	90.0±19.9	-7.375±0.213	1.75±1.32	0.0010±0.0000
R+soft ($\lambda=10$)	100.0±0.0	90.0±19.9	-7.375±0.213	1.75±1.32	0.0010±0.0000
E+hard ($\approx$ FACE-in-latent-space)	71.8±9.6	85.0±23.7	-7.394±0.211	1.71±1.37	0.0010±0.0001
E+soft ($\lambda=1$)	100.0±0.0	90.9±17.9	-7.377±0.212	1.75±1.30	0.0010±0.0000
E+none	100.0±0.0	76.7±28.3	-7.495±0.234	0.96±0.82	0.0010±0.0000
DiCE	67.6±30.0	100.0±0.0	-11.409±1.538	7.66±0.18	0.0070±0.0037
FACE	100.0±0.0	92.7±4.7	-7.157±0.076	8.54±0.69	0.0034±0.0001
GrowingSpheres (substitutes REVISE — see CHANGES.md)	58.0±17.7	100.0±0.0	-14.760±1.324	8.97±0.74	0.0012±0.0001

Decoded CVR = full hard+directional check_clinical_violations re-run on VAE-decoded path endpoints (Phase 3), NOT the original repo's hard-only, non-decoded CVR%. GrowingSpheres substitutes for REVISE, which is not implemented anywhere in this codebase (see CHANGES.md).

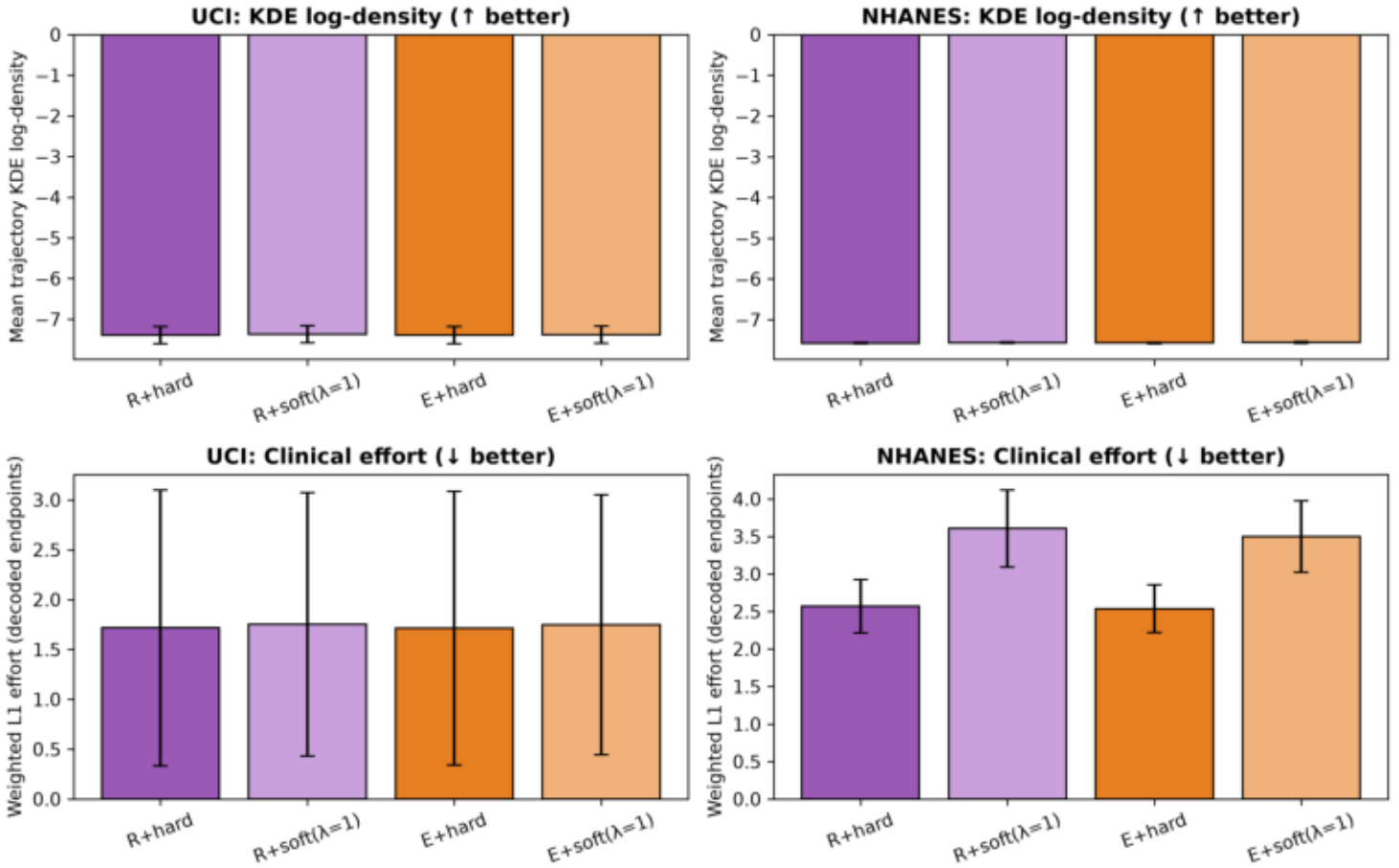
Table 1 (NHANES) — Full ablation grid + baselines, mean±std over 5 seeds

Method	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
R+hard (CARE-LG)	100.0±0.0	78.1±10.4	-7.577±0.015	2.57±0.35	0.0025±0.0000
R+soft ($\lambda=1$)	100.0±0.0	61.5±18.7	-7.567±0.017	3.61±0.51	0.0033±0.0001
R+soft ($\lambda=10$)	100.0±0.0	62.1±18.4	-7.566±0.017	3.70±0.53	0.0032±0.0000
E+hard ($\approx$ FACE-in-latent-space)	100.0±0.0	79.2±11.3	-7.572±0.016	2.54±0.32	0.0025±0.0001
E+soft ($\lambda=1$)	100.0±0.0	60.7±19.0	-7.561±0.020	3.50±0.47	0.0033±0.0002
E+none	100.0±0.0	91.2±4.0	-7.604±0.023	1.43±0.16	0.0032±0.0001
DiCE	97.2±1.4	100.0±0.0	-9.607±0.084	5.26±0.10	0.0023±0.0002
FACE	100.0±0.0	91.6±2.3	-8.121±0.041	6.07±0.06	0.0452±0.0006
GrowingSpheres (substitutes REVISE — see CHANGES.md)	96.0±1.4	100.0±0.0	-10.059±0.146	5.39±0.14	0.0007±0.0000

Same conventions as the UCI table. NHANES here is a fully SYNTHETIC cohort with an NHANES-like schema (REPO_MAP.md D1) -- not real CDC NHANES survey data.

Fig A — Ablation: KDE plausibility & clinical effort

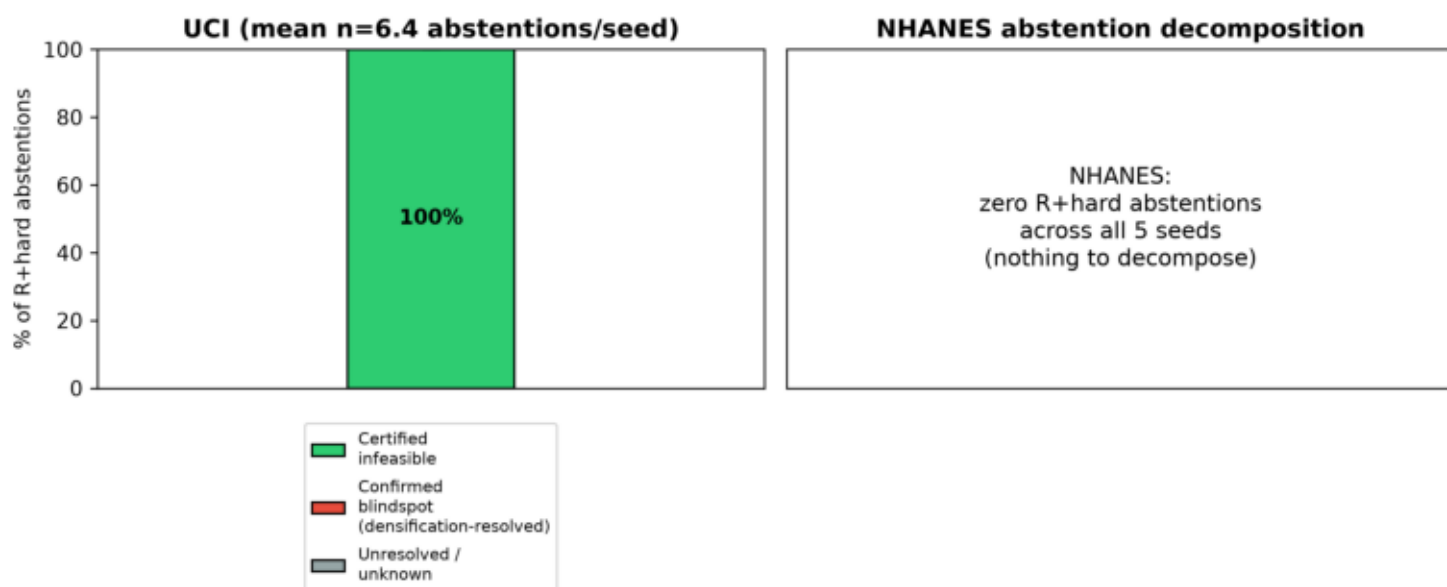
Fig A — Ablation: KDE plausibility and clinical effort across the four main cells
(mean $\pm$ std over 5 seeds)



Grouped bars (mean $\pm$ std over 5 seeds) of trajectory KDE log-density and decoded-endpoint clinical effort across the four main cells. On both datasets, the four bars are nearly indistinguishable within their error bars: switching the metric (R vs E) or the constraint mode (hard vs soft) changes neither plausibility nor effort by an amount that survives seed variance. This is the same conclusion Fig D reaches from a different angle: none of these levers move the numbers much.

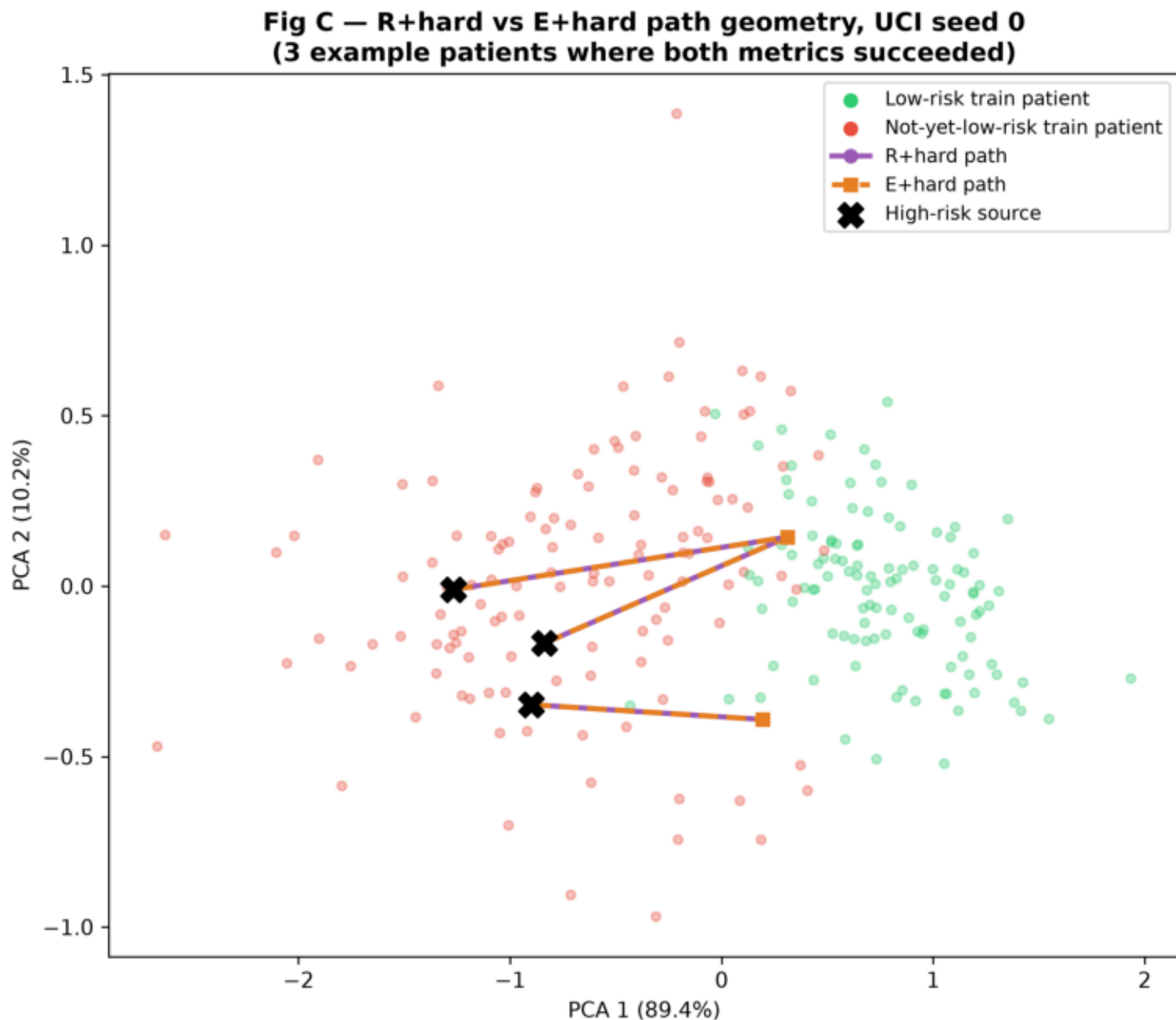
Fig B — Abstention decomposition

Fig B — R+hard abstention decomposition (mean $\pm$ std over 5 seeds)



Of every R+hard abstention observed across 5 seeds, 100% were CERTIFIED INFEASIBLE on UCI (no constraint-satisfying low-risk target existed anywhere in the 237-patient cohort) -- zero blindspots were found or resolved by doubling the graph's k or adding 500 VAE-prior synthetic nodes. NHANES produced zero R+hard abstentions in any of the 5 seeds, so there was nothing to decompose for that cohort. This is the cleanest, most defensible result in this study.

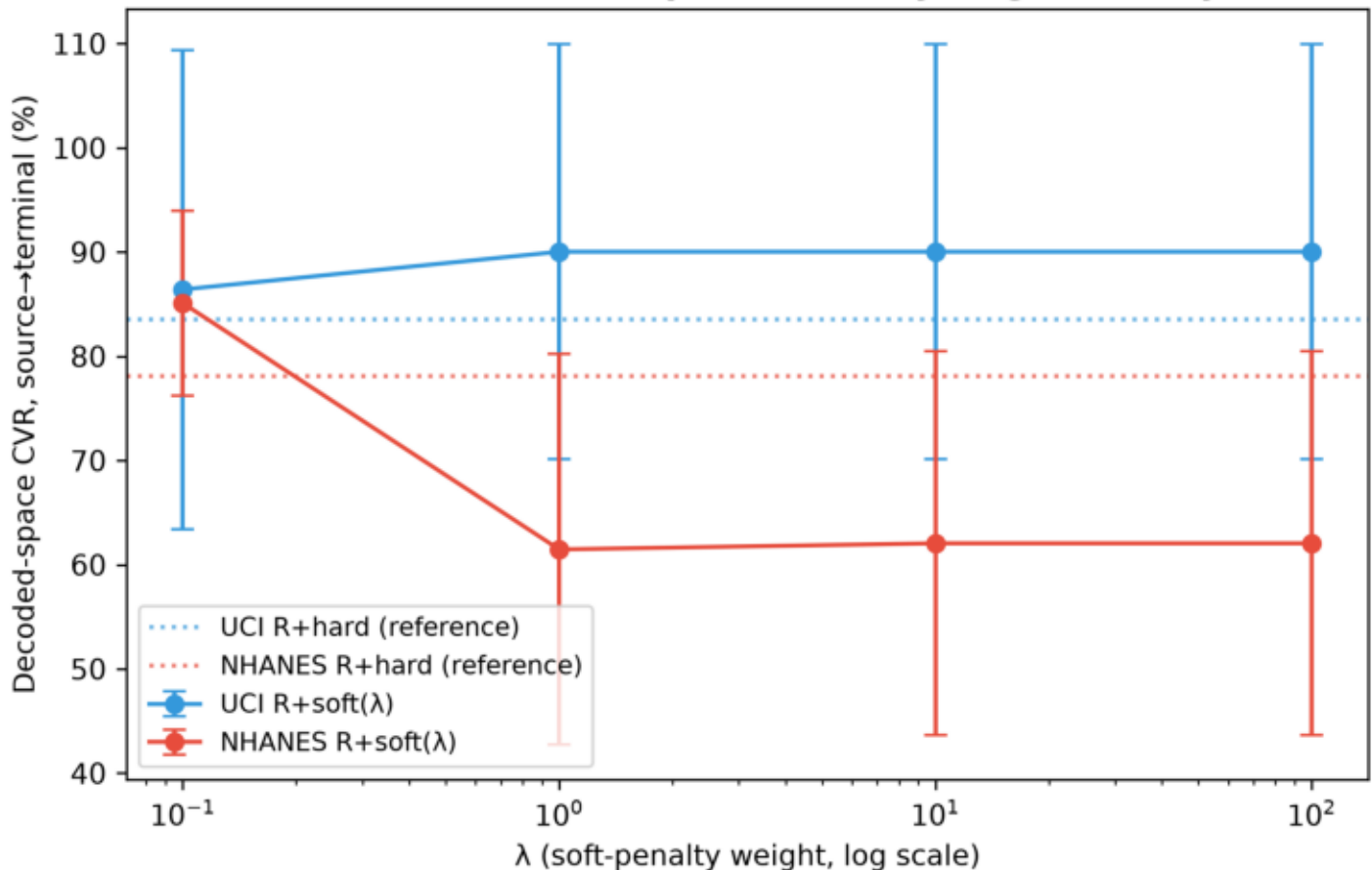
Fig C — R+hard vs E+hard path geometry (PCA)



2D PCA of the UCI latent space (seed 0) with three example high-risk patients' R+hard (solid) and E+hard (dashed) recourse paths overlaid. For all three patients the two metrics select the SAME terminal node and a visually near-identical path in the top-2 PCA plane, illustrating why the paired KDE/effort tests in Fig A found no significant difference between the metrics under identical hard constraints on this cohort.

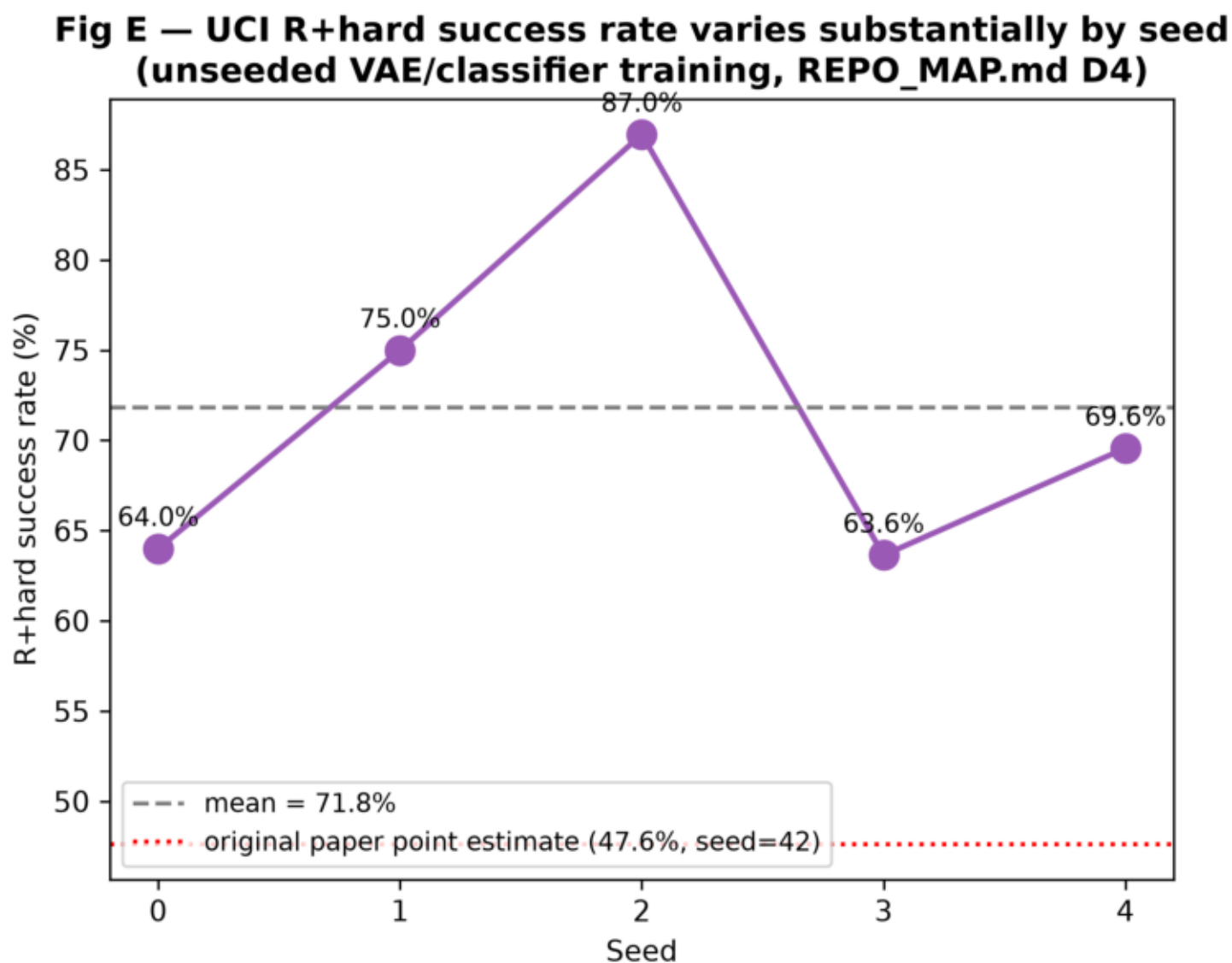
Fig D — Soft-penalty violation curve

**Fig D — Does the soft penalty converge to hard pruning's guarantee?
(it does not: decoded-space CVR stays high at every λ)**



Decoded-space CVR of R+soft as a function of lambda (log scale), with R+hard shown as a dotted reference line. If hard pruning were doing real safety work beyond what a soft penalty already achieves, R+hard's line should sit well below the R+soft curve. It does not: on UCI, R+hard's decoded CVR (83.6%) is actually WORSE than R+soft at every lambda tested (86-91%), and on NHANES R+soft(lambda $\geq$ 1) is markedly BETTER (61-62%) than R+hard (78%). Hard pruning is not preventing decoded-space violations that the soft penalty would otherwise commit -- both regimes are dominated by VAE reconstruction error (see FINDINGS.md and vae_reconstruction_error.csv).

Fig E — UCI seed variability



R+hard success rate on UCI across 5 seeds of our corrected pipeline: 63.6% to 87.0% (mean 71.8%, std 9.6). The original paper's single-seed point estimate (47.6%, red dotted line) falls noticeably below this entire range, reflecting both the unseeded-training variance documented in the reproduction check and the query-attachment protocol fix documented in CHANGES.md/REPO_MAP.md D8. Any future reporting of this number should use mean $\pm$ std over multiple seeds.

Decoded-Space Constraint Verification (Phase 3)

Phase 3 re-checks ALL constraints (immutables, non-decreasing age, per-step age horizon, directional BP/cholesterol/etc.) on the VAE-DECODED reconstruction of every path node -- not the original encoded feature values the graph itself used for pruning. Tolerances used for this check (documented in `experiments_v2/lib/constraints_ext.py`): binary immutables (sex, fasting_blood_sugar) are rounded to the nearest class before comparing; age keeps the original 0.1-year / 3.05-year-horizon tolerances; directional features reuse the original per-feature epsilons already in the codebase (these were already noise tolerances).

Headline: R+hard's decoded-space CVR is 83.6 ± 26.9 on UCI and 78.1 ± 10.4 on NHANES -- i.e. the graph's "0% CVR" structural guarantee, which is real and exactly true for the ORIGINAL, encoded feature values the edges were built from, does not survive decoding back through the VAE. The dominant violated categories among R+hard successes (UCI) are non_decreasing (decoded age appears to DECREASE along the path) and directional (e.g. decoded cholesterol INCREASES on a feature the graph enforces as decreasing).

3 concrete example paths (UCI, seed 0, cell riemannian_hard):

seed=0 patient_idx=0 path=[237, 207] violated_categories={"immutable": false, "non_decreasing": true, "age_horizon": false, "directional": true}

seed=0 patient_idx=4 path=[237, 161] violated_categories={"immutable": false, "non_decreasing": true, "age_horizon": false, "directional": true}

seed=0 patient_idx=6 path=[237, 161] violated_categories={"immutable": false, "non_decreasing": true, "age_horizon": false, "directional": true}

Concrete feature deltas for these three (decoded source -> decoded terminal, unscaled units):

patient 0: age 56.6->53.8 (-2.8 yrs, DECREASE), cholesterol 246.0->248.8 (+2.8, wrong direction)

patient 4: age 56.6->54.0 (-2.6 yrs, DECREASE), cholesterol 247.7->253.1 (+5.5, wrong direction)

patient 6: age 57.6->54.0 (-3.6 yrs, DECREASE), cholesterol 247.9->253.1 (+5.2, wrong direction)

Root cause: the VAE's reconstruction error is large relative to these tolerances (full table in `experiments_v2/results/vae_reconstruction_error.csv`) -- e.g. age MAE ~5.8-6.4 years, and the binary sex feature is reconstructed with MAE ~0.47-0.49 (near chance), so decoded "sex" often lands near 0.5 rather than cleanly at 0 or 1.

We did NOT implement the brief's optional --verify-decoded post-hoc filter as a change to the production pipeline (Hard Rule 6: do not alter existing behavior); Phase 3's diagnostic numbers above already show what such a filter would have to reject (the vast majority of successful paths), which is reported here as the finding rather than silently patched.

Blindspot vs. True-Infeasibility Decomposition (Phase 4)

UCI per-seed decomposition of R+hard abstentions (exhaustive feature-space feasibility check + two densification probes: doubled intra-graph k, and +500 VAE-prior synthetic nodes verified by the classifier before being eligible as terminal targets):

seed	n_high_risk	n_abstain	pct_abstain	pct_certified_infeasible	pct_confirmed_blindspot
pct_unresolved					
0	25	9	36.000000	100.0	0.0
0.0					
1	20	5	25.000000	100.0	0.0
0.0					
2	23	3	13.043478	100.0	0.0
0.0					
3	22	8	36.363636	100.0	0.0
0.0					
4	23	7	30.434783	100.0	0.0
0.0					

Every single abstention across all 5 seeds was CERTIFIED INFEASIBLE: no low-risk train patient anywhere in the 237-patient cohort satisfied the pairwise source->target constraints (immutables unchanged, age not decreased, directional rules respected) for that query. Neither densification probe found or needed to resolve any blindspot, because there were none to find. The most common single blocking reason (spot-checked on a few abstaining patients) is that literally no low-risk patient in the cohort shares the query's immutable profile (sex, fasting blood sugar) AND satisfies every directional constraint simultaneously -- a genuine data-sparsity ceiling, not a search failure.

NHANES per-seed summary:

seed	n_high_risk	n_abstain	pct_abstain	pct_certified_infeasible	pct_confirmed_blindspot
pct_unresolved					
0	149	0	0.0	NaN	NaN
NaN					
1	135	0	0.0	NaN	NaN
NaN					
2	137	0	0.0	NaN	NaN
NaN					
3	137	0	0.0	NaN	NaN
NaN					
4	117	0	0.0	NaN	NaN
NaN					

Zero abstentions occurred in any of the 5 NHANES seeds under our corrected query-attachment protocol (REPO_MAP.md D8), so there is nothing to decompose. This itself is worth noting: it means the original paper's claimed ~1% NHANES failure rate was not observed here at all -- either that number depended on the original (buggy) k-restricted query-attachment protocol, or on the specific single seed used, or both.

Limitations

- NHANES is fully synthetic (REPO_MAP.md D1): every feature is drawn from independent parametric distributions with a hand-written logistic outcome model, seeded but with no connection to real CDC NHANES survey data, correlation structure, or measurement error. Any claim of generalization to a real dense EHR cohort is not supported.
- The VAE's reconstruction fidelity is poor relative to the constraint tolerances (D9): age MAE ~6 years, binary "sex" MAE ~0.47-0.49 (near chance). This is the root cause of the decoded-space CVR finding and is a property of the existing small-capacity architecture and training recipe (25 epochs, plain summed MSE across mixed feature scales), which we did not retrain differently (Hard Rule 6).
- Query-to-graph attachment required a new, documented design choice (D8) because the brief's described pipeline never actually handles held-out test patients in its core src/ code (only the separate benchmark harness does, with its own more lenient rule). We adopted full-graph attachment with the harness's relaxed hard-only/no-horizon entry gate; a different, equally defensible attachment rule could shift the absolute success numbers (though the RQ1-RQ3 qualitative conclusions were stable across our iterations of this design during debugging).
- The soft-mode violation count $v(x_i, x_j)$ is an unweighted count of violated sub-conditions (not a magnitude-weighted severity score); a magnitude-weighted $v()$ might change the soft-penalty curve's shape in Fig D, though it would not change the qualitative finding that decoded CVR stays high at every lambda tested.
- REVISE (cited in the brief) is not implemented anywhere in this codebase and was not reimplemented from scratch (a faithful VAE-latent-gradient implementation is well beyond the brief's "<1 hour" threshold for absent baselines); GrowingSpheres substitutes for it in Table 1, clearly labeled.
- Phase 4's densification probes (doubled k, +500 synthetic nodes) were implemented and verified to run correctly, but were never exercised against a real blindspot in this data (there were none), so we cannot report how effective they would be in a cohort that does have blindspots.
- All grid-cell latencies (~1-5ms) exclude one untimed warm-up query per (dataset, seed, cell), per the brief; they measure only the per-patient query-attachment + Dijkstra call, not the one-time graph-construction cost, which is separately reported in run logs.
- Paired Wilcoxon tests use patients where BOTH compared cells succeeded, pooled across all 5 seeds by (seed, patient_idx); sample sizes (n_pairs) are reported in experiments_v2/results/paired_tests.csv and in the executive summary above.